

Project 3

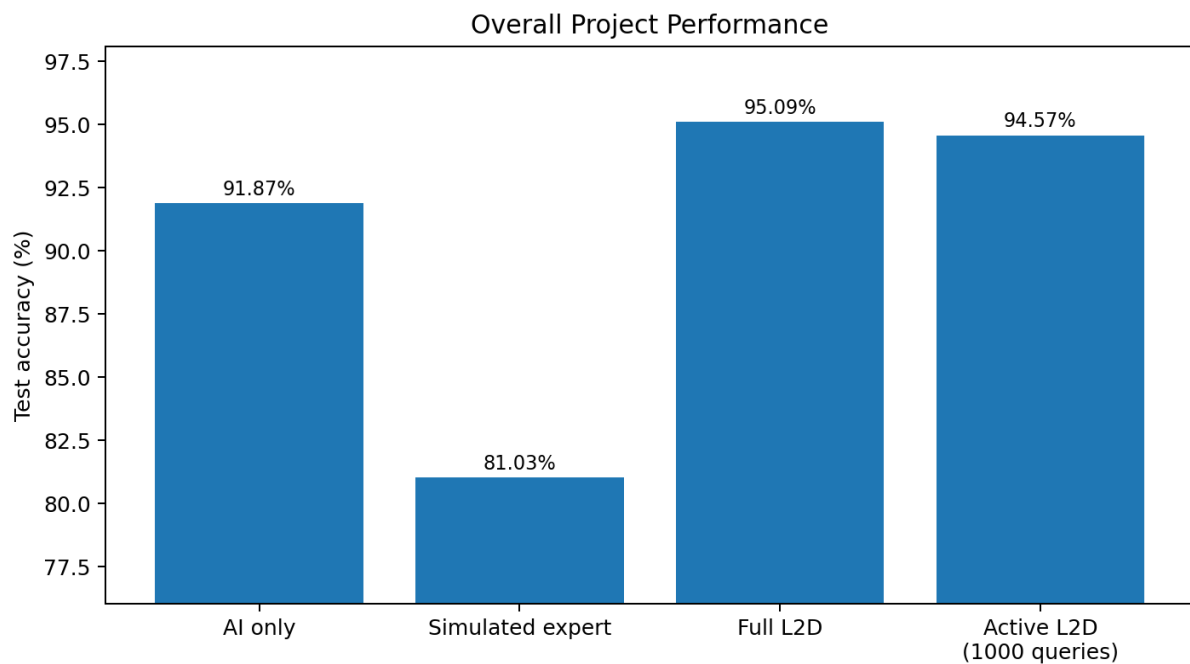
Active Learning for Learning-to-Defer

Human-Centric Artificial Intelligence

This report describes the experiments performed for Project 3, provides justification for the main design choices, and presents a detailed analysis of the experimental results.

The project studies human-AI collaboration through Learning-to-Defer and Active Learning using the AG News topic-classification dataset.

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1. Project Overview

The objective of the project is to train a classifier capable of collaborating with an expert. Instead of always producing its own prediction, the system may defer selected examples to the expert when expert involvement is expected to improve the final decision.

AG News is used as the classification task. Articles belong to four categories: World, Sports, Business, and Sci/Tech. The complete labeled training split is used to train the baseline classifier. The official test split is reserved exclusively for evaluation.

Experimental Progression

The experiments are organized progressively. Task 1 establishes an AI-only baseline. Task 2 introduces an imperfect simulated expert with specialized competence. Task 3 assumes complete expert information and learns when to defer. Task 4 removes access to expert labels and uses Active Learning to determine which expert predictions should be queried.

System	Test accuracy
AI baseline	91.87%
Simulated expert	81.03%
Full-information L2D	95.09%

2. Task 1 - Baseline Classification Model

Task Objective

The first experiment trains a classifier on all available AG News training labels and evaluates its performance on the held-out test set. This provides the AI-only reference performance that the later human-AI system should aim to match or improve.

Design Choice and Justification

The text representation is **TF-IDF** and the classifier is **Logistic Regression**. TF-IDF provides an efficient sparse representation of news text and captures discriminative unigram and bigram information. Logistic Regression is computationally efficient for high-dimensional sparse text features and provides class-probability estimates. These probabilities are especially useful later in Learning-to-Defer because the maximum predicted probability is used as an estimate of AI confidence.

Experimental Setup

The classifier is trained on all **120000** AG News training examples. The test set contains **7600** examples and is used only after training for evaluation.

Results

Metric	Value
Training samples	120000
Test samples	7600
Test accuracy	91.87%

Class	Precision	Recall	F1-score	Support
World	0.93	0.91	0.92	1900
Sports	0.95	0.98	0.97	1900
Business	0.89	0.88	0.89	1900
Sci/Tech	0.89	0.90	0.90	1900

Confusion Matrix

Actual / Predicted	World	Sports	Business	Sci/Tech
World	1726	53	73	48
Sports	20	1860	12	8
Business	56	17	1677	150
Sci/Tech	47	19	115	1719

Result Analysis

The baseline reaches **91.87%** test accuracy. Sports is the strongest class, while the largest confusion occurs between Business and Sci/Tech. This weakness motivates the design of a complementary simulated expert with particular competence in those categories.

3. Task 2 - Simulated Expert

Task Objective

Task 2 introduces an imperfect simulated expert. The expert should not simply reproduce the classifier or behave perfectly; instead it should exhibit strong performance in specific regions of the input space and weaker performance elsewhere.

Expert Design and Justification

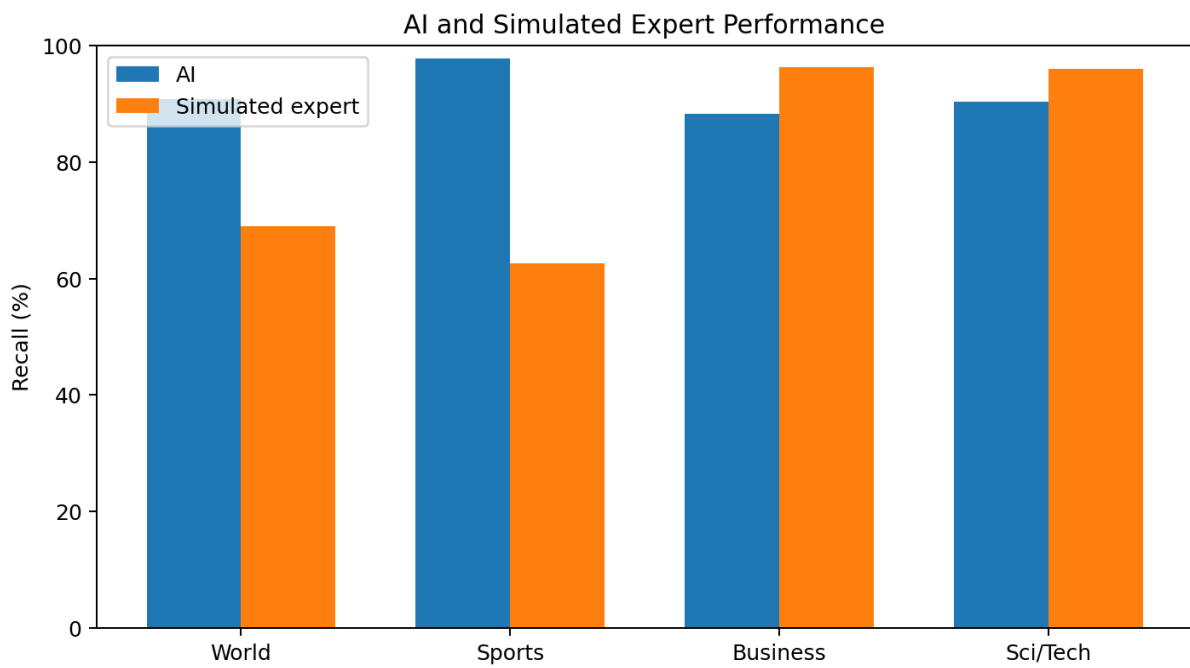
The simulated expert is modeled as a **Business & Technology Specialist**. The expert is intentionally highly competent on Business and Sci/Tech articles and weaker on World and Sports. This choice is motivated by the Task 1 results, where Business and Sci/Tech were the most difficult categories for the AI. The resulting complementary error patterns make the expert useful for studying selective deferral.

Class	Designed expert accuracy
World	70.00%
Sports	65.00%
Business	97.00%
Sci/Tech	96.00%

Results

Predictions are sampled from the predefined class-dependent competence profile using a fixed random seed, which makes the experiment reproducible. On the AG News test set the simulated expert achieves an overall accuracy of **81.03%**.

Class	Precision	Recall	F1-score	Support
World	0.79	0.69	0.74	1900
Sports	0.89	0.63	0.74	1900
Business	0.78	0.96	0.86	1900
Sci/Tech	0.81	0.96	0.88	1900



Strengths and Weaknesses

The expert is substantially stronger on Business and Sci/Tech, while the AI is much stronger on World and Sports. Although the expert's overall accuracy is below the AI-only accuracy, the complementary competence profile creates useful opportunities for collaboration. This is essential for Task 3 because deferral is beneficial only when the expert is more reliable than the AI for a particular input.

4. Task 3 - Learning to Defer

Task Objective

Task 3 assumes that both the true class labels and simulated expert predictions are available during training. The objective is to learn whether each new input should be handled by the AI classifier or deferred to the expert.

Deferral Strategy and Design Choice

A separate competence classifier is trained to estimate $P(\text{expert correct} \mid \mathbf{x})$. The training target is 1 when the simulated expert prediction matches the true AG News label and 0 otherwise. For a test article, the estimated expert competence is compared with the maximum class probability produced by the baseline classifier. The article is deferred when estimated expert competence is greater than AI confidence.

This strategy was selected instead of a simple AI-confidence threshold because low AI confidence alone does not imply that the expert will perform better. The competence model explicitly incorporates the expert's observed strengths and weaknesses.

Evaluation Metrics

The evaluation considers overall human-AI team accuracy as well as deferral quality. Coverage measures the proportion handled by the AI, while the deferral rate measures expert workload. Useful deferrals are cases where the AI would be wrong and the expert is correct. Harmful deferrals are cases where the AI would be correct but the expert is wrong. Routing accuracy measures whether the system chooses the better agent when exactly one of them is correct.

Metric	Result
AI-only accuracy	91.87%
Expert-only accuracy	81.03%
Human-AI team accuracy	95.09%
Coverage	82.17%
Deferral rate	17.83%
Routing accuracy	84.58%
Useful deferrals	314
Harmful deferrals	69
Oracle upper bound	98.95%

Result Analysis

The combined system reaches **95.09%** accuracy, improving the AI baseline by **3.22 percentage points**. The model defers **17.83%** of test examples rather than sending all examples to the expert.

There are **314** useful deferrals and only **69** harmful deferrals. The routing accuracy of **84.58%** shows that the learned router usually selects the better decision maker when AI and expert capabilities differ.

Class-wise Deferral Behaviour

Class	Samples	Deferred	Deferral rate	Team accuracy
World	1900	144	7.58%	92.26%

Class	Samples	Deferred	Deferral rate	Team accuracy
Sports	1900	61	3.21%	97.89%
Business	1900	572	30.11%	94.68%
Sci/Tech	1900	578	30.42%	95.53%

The substantially larger deferral rates for Business and Sci/Tech demonstrate that the competence model has learned the intended expert specialization rather than applying a uniform deferral rule.

5. Task 4 - Active Learning for Expert Competence Discovery

Task Objective

Task 4 removes the assumption that expert predictions are available during training. The baseline classifier still has access to the complete labeled AG News training set, but the competence learner starts without expert data. It may reveal an expert prediction only by actively querying a selected training example.

Active Learning Design Choice

The primary strategy is pool-based **competence uncertainty sampling**. After an initial class-balanced query set, a competence model is trained from the expert observations collected so far. The acquisition function prioritizes examples for which $P(\text{expert correct} | \mathbf{x})$ is closest to 0.5. These are the samples on which the current competence model is least certain.

Random querying is included as a passive baseline so that the value of actively selecting queries can be measured at equal annotation budgets. An additional deferral-aware diversity strategy was evaluated. It prioritizes examples near the AI-expert routing boundary and uses TF-IDF cosine similarity to reduce redundant queries.

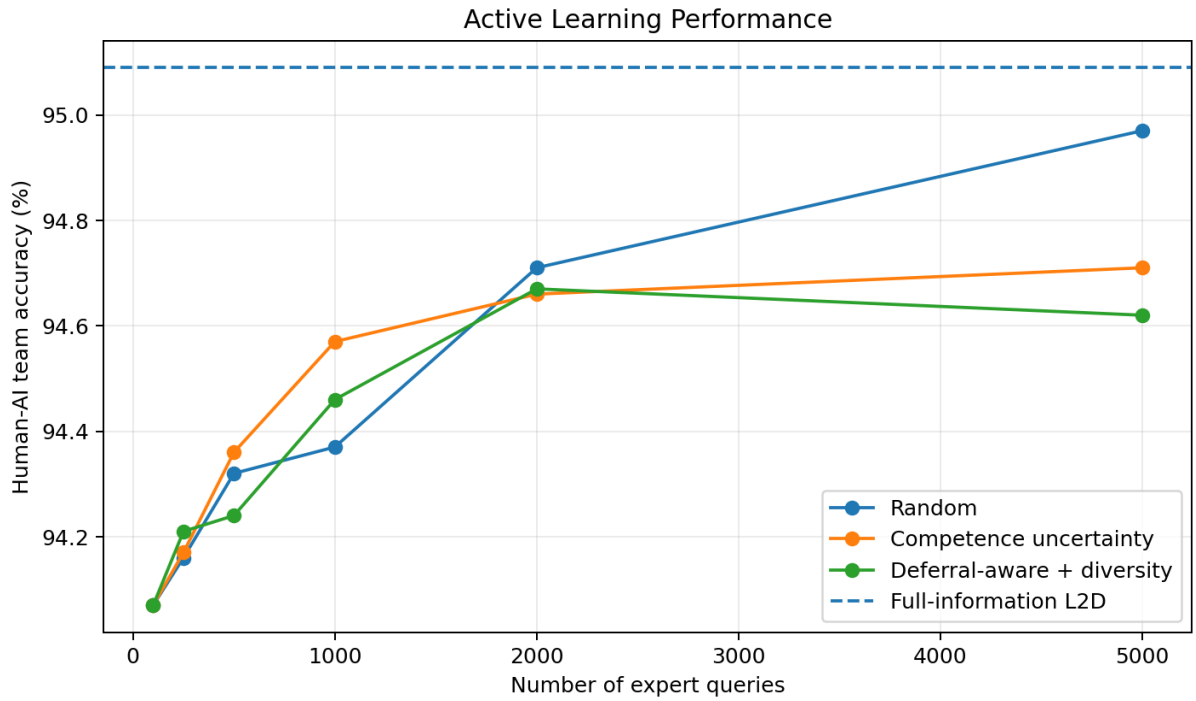
Experimental Protocol

All strategies begin with the same 100 class-balanced expert queries and are evaluated at query budgets of 100, 250, 500, 1000, 2000, and 5000. At each budget, the learned competence model is combined with the same baseline AI classifier and Task 3 routing rule. This ensures that differences in performance are attributable to the expert-query strategy.

Evaluation Metrics

The main metric is human-AI team accuracy. Routing accuracy measures the quality of AI-versus-expert selection, while deferral rate measures the amount of expert involvement. The experiment also reports the fraction of the full-information Task 3 accuracy improvement recovered with each limited expert-query budget.

Queries	Train %	Random	Uncertainty	Deferral + diversity
100	0.08%	94.07%	94.07%	94.07%
250	0.21%	94.16%	94.17%	94.21%
500	0.42%	94.32%	94.36%	94.24%
1000	0.83%	94.37%	94.57%	94.46%
2000	1.67%	94.71%	94.66%	94.67%
5000	4.17%	94.97%	94.71%	94.62%



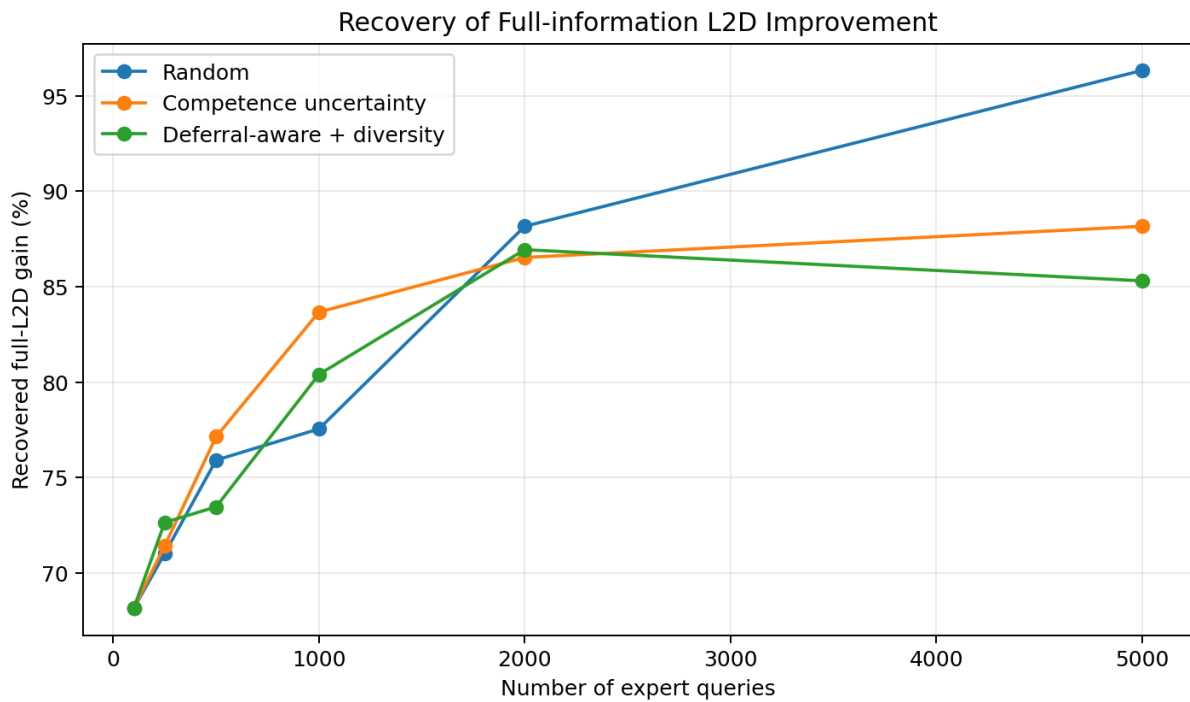
Key Active Learning Result

Metric	Result
Expert queries	1000
Training set queried	0.83%
Team accuracy	94.57%
Routing accuracy	82.47%
Recovered full-L2D gain	83.67%
Gap to full-information L2D	0.52 percentage points

Result Analysis

At the selected low-budget operating point, only **1000** of the 120,000 training examples are queried. This corresponds to approximately **0.83%** of the training dataset.

Using these expert queries, competence uncertainty sampling reaches **94.57%** team accuracy with a routing accuracy of **82.47%**. This recovers **83.67%** of the accuracy improvement achieved when all expert labels are available in Task 3.



Comparison with Random Querying

Competence uncertainty sampling provides its clearest benefit at small and medium query budgets. At larger budgets, random sampling becomes competitive. At 5000 expert queries, random querying reaches **94.97%** team accuracy, while competence uncertainty reaches **94.71%**.

This result suggests that targeted Active Learning is particularly valuable when expert interaction is expensive and the query budget is strongly limited. When many labels can be requested, random sampling eventually obtains broad coverage of the relatively simple expert competence regions.

6. Task 5 - Active Learning with Human Expert

Interactive Extension

Task 5 replaces the simulated interaction with a real user interface. Articles selected by the active-learning procedure are presented to the user one at a time. The user assigns one of the four AG News labels while the ground-truth label remains hidden.

Design Choice

The interface uses articles selected through competence uncertainty sampling. This connects the human interaction directly to the Active Learning objective: requesting labels only for examples considered informative for expert competence discovery.

Metric	Result
Human labels provided	6
Correct labels	6
Incorrect labels	0
Human accuracy	100.00%

These results are reported separately from the controlled simulated-expert experiments in Tasks 1-4. Human responses therefore do not overwrite the reproducible Task 1-4 results.

7. Overall Discussion

The four experiments demonstrate the importance of modeling complementary human and AI capabilities rather than considering standalone accuracy alone. Although the simulated expert achieves only **81.03%** overall accuracy, its specialization allows the full-information human-AI system to improve from **91.87%** AI-only accuracy to **95.09%**.

The Task 3 results confirm that learning expert competence is useful for routing decisions. Task 4 then shows that this competence profile does not need to be learned from every possible expert prediction. Active Learning can recover a substantial fraction of the full-information benefit using only a small expert-query budget.

8. Conclusion

The project successfully implements a complete progression from standard classification to human-AI collaboration and Active Learning.

The AI-only classifier achieves **91.87%** accuracy. The simulated expert achieves **81.03%**, but provides complementary expertise. Learning-to-Defer combines these capabilities to reach **95.09%** accuracy.

With Active Learning, competence uncertainty sampling reaches **94.57%** using only **1000** expert queries (0.83% of the training set), recovering **83.67%** of the full-information Learning-to-Defer improvement.